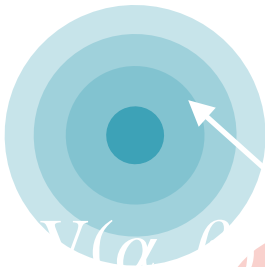
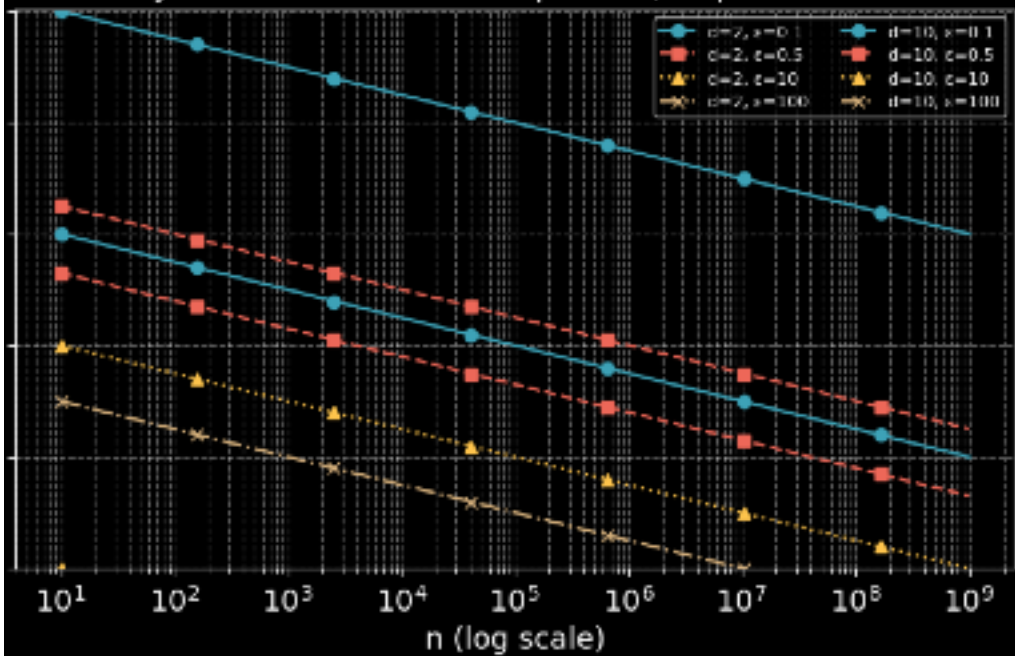


$1/\sqrt{n}$, independent of d

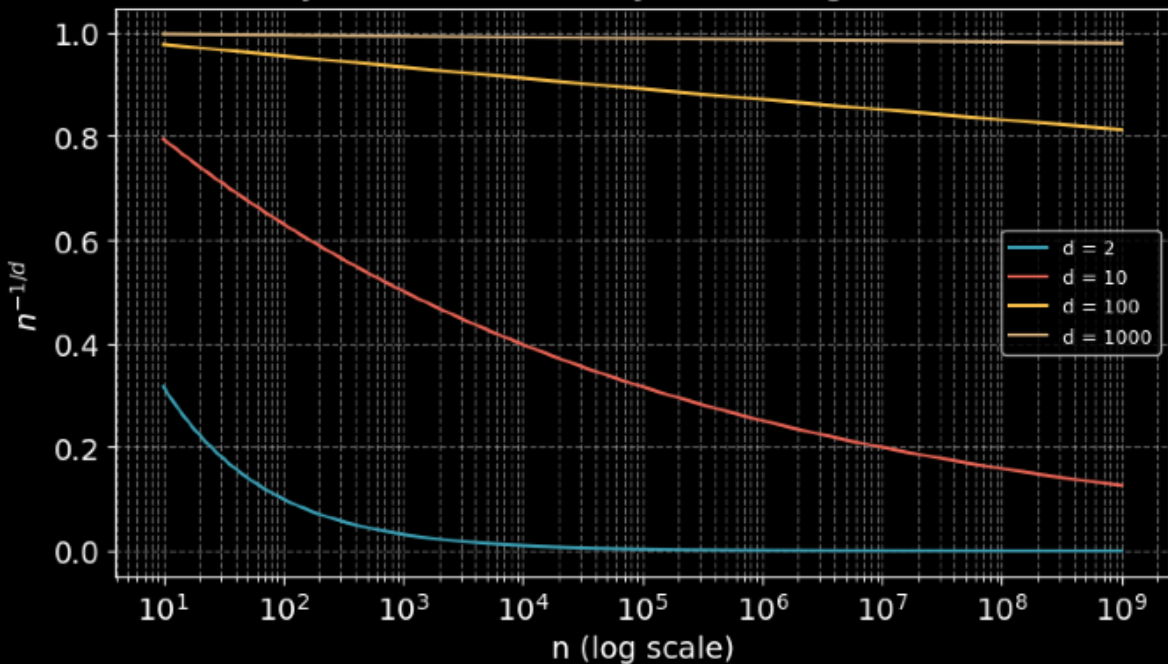
it also improves statistical efficiency



Decay of $\varepsilon^{-d/2} \cdot n^{-1/2}$: d-independent, dependence on ε

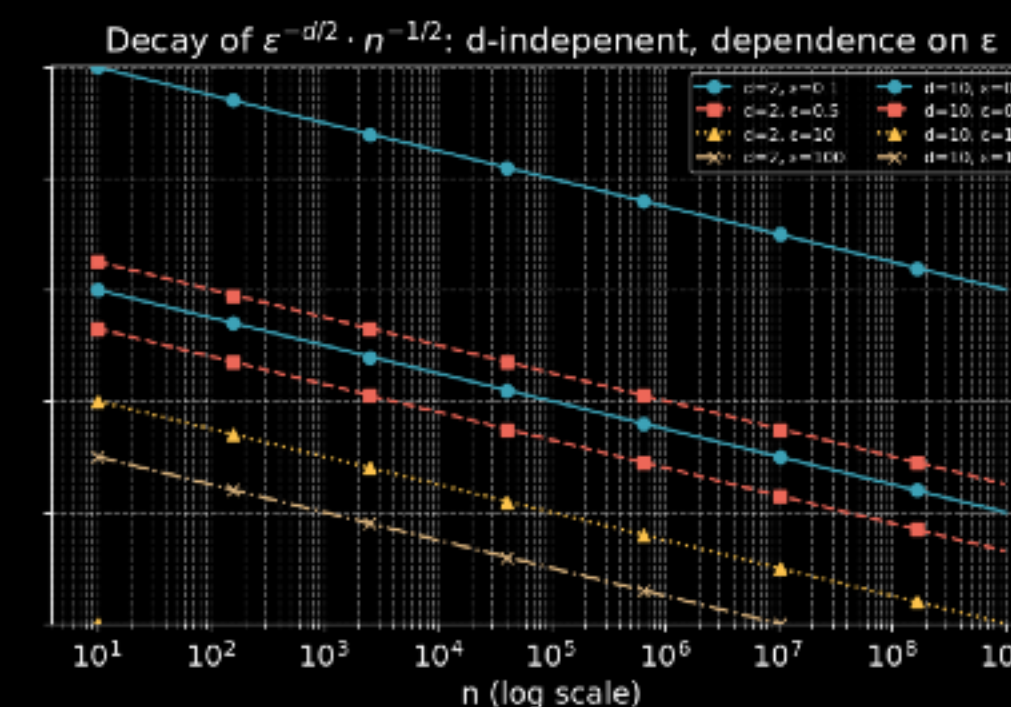
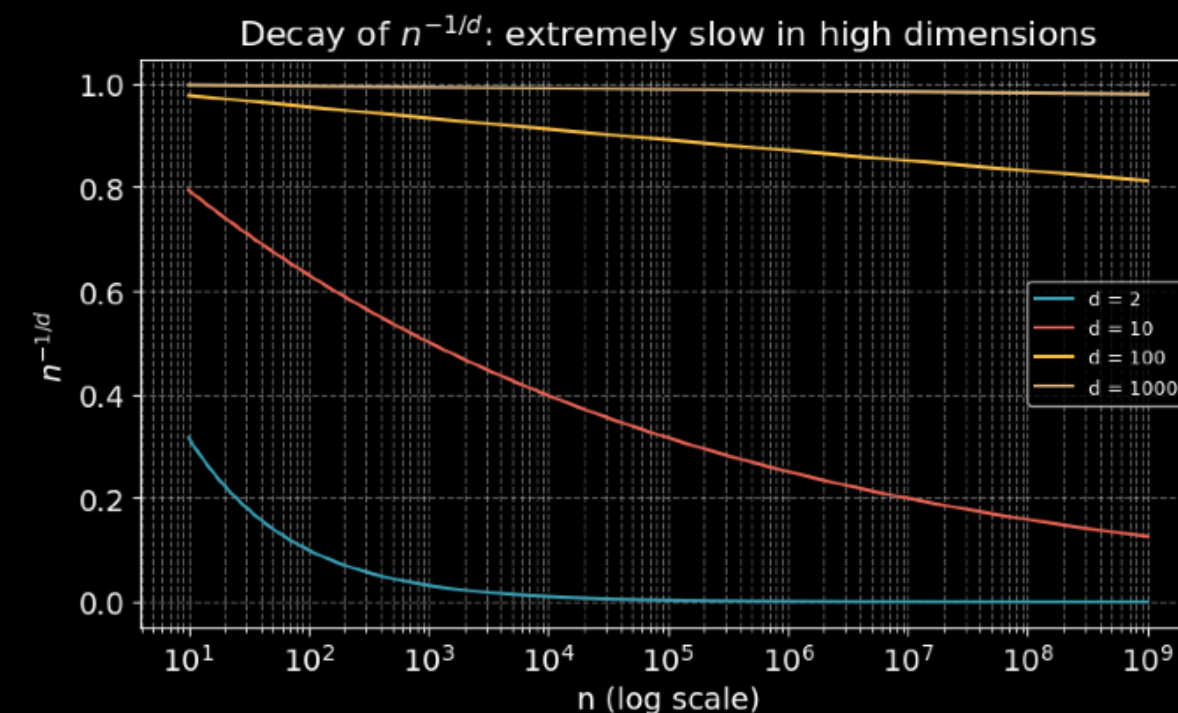
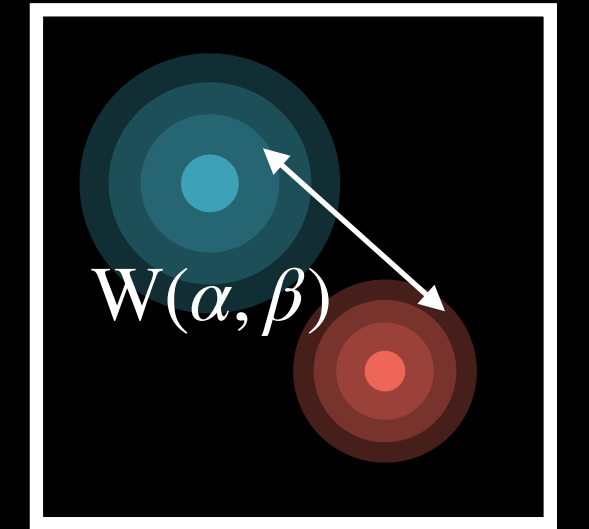
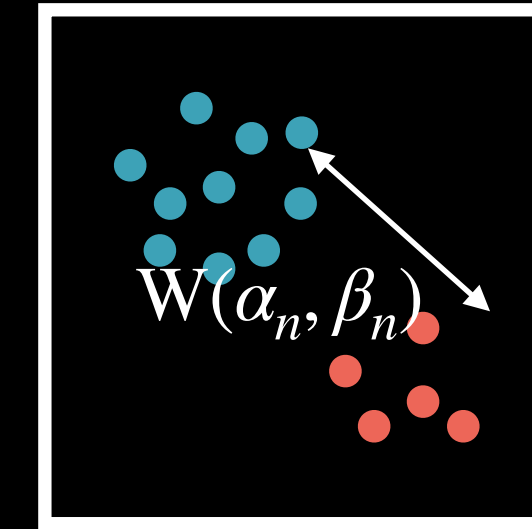


Decay of $n^{-1/d}$: extremely slow in high dimensions



Sample Complexity of SD

- Classic OT: $\mathbb{E}(|W_p(\hat{\alpha}_n, \hat{\beta}_n) - W_p(\alpha, \beta)|) = O(n^{-\frac{1}{d}})$
- But, with entropic regularization:
 - $\mathbb{E}(|W_{\varepsilon,p}(\hat{\alpha}_n, \hat{\beta}_n) - W_{\varepsilon,p}(\alpha, \beta)|) = O(\varepsilon^{-\frac{d}{2}} n^{-\frac{1}{2}}) \xrightarrow{\varepsilon \rightarrow \infty} O(n^{-\frac{1}{2}})$
 - Rate on n : $1/\sqrt{n}$, **independent of d** ; the ‘best’ achievable rate
 - Empirically much more stable in high-dim ML tasks.



- **Takeaway:**
 - entropy not only speeds up compute, **it also improves statistical efficiency.**

Multi-Faceted Interpolation

Classic OT

Sinkhorn Divergence

MMD

Computational
Complexity

Statistical
Complexity

Coupling

Geometry